Biochemistry Cume – February 13, 2015

1 (50 pts). The SDS-PAGE of a protein yields two bands corresponding to molecular masses of 10 and 17 kD. After cross-linking this protein with dimethyl suberimidate under sufficient dilution to eliminate intermolecular cross-linking, SDS-PAGE of the product yields 10 bands with molecular masses 10, 17, 27, 30, 34, 44, 51, 61, 71, and 81 kD. Assuming that dimethyl suberimidate can cross-link only contacting subunits, diagram the quaternary structure of the protein.

2 (25 pts). A molecule of amylopectin consists of 1000 glucose residues and is branched every 30 residues. How many reducing ends does it have?

3 (25 pts). Amino acid compositions of polypeptides with unknown sequences are written in parentheses with commas separating amino acid abbreviations such as in (Gly, Tyr, Val). Known amino acid sequences are written with residue names in order and separated by hyphens; for example, Tyr-Val-Gly.

Treatment of a polypeptide by 2-mercaptoethanol yields two polypeptides that have the following amino acid sequences:

1. Ala-Lys-Cys-Met-Arg-Cys-Ala-Arg-Cys-Asn
2. Val-Cys-Arg-Val-Ile-Lys-Gly-Ala-Lys

Trypsin-catalyzed hydrolysis of the intact polypeptide yields polypeptide fragments with the following amino acid compositions:

3. (Ala, Lys)
4. (Cys, Val, Met, Arg)
5. (Cys, Ala, Arg, Asn)
6. (Ile, Lys, Val)
7. (Gly, Ala, Lys)

Indicate the positions of the disulfide bonds in the original polypeptide.